

Week 1: Urban Data Integration Platform

1. Purpose and data catalog

The platform turns four independently supplied datasets into standardized Delta tables and a trip-level analytical table. A implements reusable ingestion and storage (Tasks 2-3), B defines the data catalog and cleaning contract (Tasks 1 and 4), and C implements integration and storage experiments (Tasks 5-6). Week 1 uses reproducible full-batch processing on a local machine.

Taxi Trips

Entity: one yellow-taxi journey. There is no supplied trip ID, so the key is a SHA-256 fingerprint of 19 business fields; numeric measurements are formatted to six decimals. Pickup/dropoff location IDs join to zones; pickup UTC hour joins to environmental observations. Temporal fields include both trip timestamps, local pickup date and duration. Vendor, payment, rate and location codes are categorical. Trip rows and ingestion batches grow over time. The supplied January-March 2024 files contain 9,554,778 rows.

Weather

Entity: one hourly background-weather record. The unique key is the source year/month/day/hour, represented as `weather_hour_utc` for joins. Condition/cloud codes and measurement-source labels are categorical. Hourly rows grow over time. There are 8,784 records for 2024. The file has no city, station or timezone metadata: NYC background and UTC are explicit project assumptions, not facts established by matching rates.

Air Quality

Entity: one EPA monitor-hour PM2.5 observation. The key comprises state, county, site, pollutant, POC, GMT hour and method code. GMT hour is the temporal join key; site/county/location fields retain spatial context. Pollutant, method, qualifier and site codes are categorical. Monitor-hours and station inventories can grow. The national file has 8,139,551 rows; the agreed NYC five-county scope contains 51,885 rows from six sites, located in Bronx, Kings and Queens.

Taxi Zone Lookup

Entity: one taxi zone, keyed by `location_id` (1-265). This ID supports both pickup and dropoff joins. Borough, zone and service-zone labels are categorical; the snapshot has no temporal field. It changes slowly through additions or renaming. IDs 264 and 265 represent Unknown and Outside of NYC. All 265 supplied rows are retained.

Detailed field inventory and verified observations: `docs/data_catalog.md` and `docs/data_contract.md`.

2. Storage and reusable ingestion

Storage layout

Original files remain under `data/raw/`. Each leaf under `data/delta/standardized/` and `data/delta/rejected/` is a separate Delta table. `metadata/ingestion_runs` stores execution records; `integrated/integrated_taxi_trips` stores enriched trips. Experiments use fresh `data/benchmark/run_id/round_n/S0` or `S1` directories. Columns and table names use `snake_case`; UTC timestamps use the `_utc` suffix. Delta combines Parquet data files with a transaction log and is read through its table directory.

Taxi Zones is the lookup table. Weather, NYC Air, Zones, rejected rows and metadata remain unpartitioned because their current sizes do not justify extra directories and files. Standardized Taxi is the unpartitioned baseline. Integrated Taxi initially uses New York pickup_date partitions; this choice remains provisional. Task 6 compares unpartitioned Taxi with daily partitions using identical inputs and writer settings.

Partitioning can be harmful with small tables, many small partitions, skew, or queries that scan all dates. With 20 times more data, reassess daily versus monthly layouts and target file sizes, compact small files and introduce incremental writes. Small reference tables need not be partitioned merely because Taxi grows. Distributed storage would require URI-aware path handling and connectors; the current implementation is local.

Common pipeline

Configuration -> `read_source()` -> `prepare()` -> `accepted/rejected/metrics` -> `write_delta()` -> readback verification -> ingestion metadata.

The reader handles CSV and Parquet, validates CSV headers against explicit schemas, and checks required source columns. `prepare()` applies registered transformations and validation, classifies duplicates and returns a common result. A disk-only intermediate cache prevents the wide Taxi audit rows from exhausting the local JVM heap; it is released after writing.

Source paths, versions, timezones and scope are configured in `configs/datasets.json`. Schemas, transformers and validation rules remain dataset-specific because field semantics differ. Shared orchestration, writing and metadata avoid copying a complete pipeline for every source. Twenty new datasets would require corresponding configuration/schema/transformer/validator registrations, CLI batch registration and tests, with separate integration rules where appropriate.

Metadata records counts, exclusions, rejections, duplicate/quality codes, versions, UTC execution times and status. A `completed_batch.json` handoff is published only after all four tables succeed and records their Delta versions. Integration reads those versions. Single-table reruns invalidate the handoff; this local workflow assumes one ingestion writer.

3. Common model and quality rules

Naming, types and time

Names are normalized to snake_case; normalization collisions are rejected. CSV types are declared in schemas.py; Taxi retains the supplied Parquet types. Canonical times are Spark timestamp values in a UTC session. Source times and fields are retained for audit. Every accepted/rejected row carries source_file, run_id, schema_version and rule_version; rejected rows also retain raw_record_json and error_reasons.

Taxi PU/DO fields become pickup_location_id and dropoff_location_id. Pickup/dropoff timestamps are interpreted as New York wall times and converted to UTC. pickup_date remains the New York calendar date; pickup_hour_utc is the truncated UTC hour. Duration is the difference between converted instants, including daylight-saving changes. Other business values are retained. The fingerprint excludes batch/file context; identical canonical business records are treated as duplicates.

Weather combines year/month/day/hour, parses the source timestamp and applies its configured timezone. Measurements and their source labels remain unchanged, without imputation or unit conversion. Units cannot be independently verified from the supplied metadata. snwd and wpgt are entirely missing; optional nulls remain null.

Air first filters the agreed state/counties, keeping national raw data. GMT date/time is authoritative; local-standard date/time is parsed separately and is not treated as New York daylight-saving time. Sample Measurement and Units of Measure are renamed. State/county/site codes are padded to 2/3/4 characters, parameter/method codes are strings, and blank qualifiers become null. Site/monitor IDs and the compound fingerprint are derived; remaining source fields are preserved.

Zone text is trimmed. IDs 264/265 receive their explicit special labels for borough, zone and service_zone; the zone ID is hashed as record_id. The complete transformation inventory is maintained in docs/data_contract.md.

Rejection versus flags

Missing keys, invalid required times, negative distances/passenger counts, invalid zone IDs, configured weather-domain violations, NaN/Infinity measurements, wrong Air pollutant/unit and duplicate keys are rejected. Taxi pickups must lie within the supplied January-March period, and dropoff cannot precede pickup. Duplicate ranking prefers a valid row, then source file and raw JSON for deterministic selection.

Potential refunds, zero-distance/duration trips, extreme durations/distances, optional weather gaps and EPA qualifiers are flagged rather than silently deleted. Non-NYC Air records are scope exclusions, not quality failures. Within scope, input = accepted + rejected; for Air, national input = excluded + scoped input.

4. Integration, validation and trade-offs

One row per accepted trip

Taxi is left-joined twice to the unique 265-row zone lookup, adding pickup/dropoff zone and borough names. The small lookup is broadcast. Missing or special locations retain the original trip; right-hand key uniqueness is checked before joining, rather than hiding join duplication afterwards.

Weather is already unique by hour. Air is restricted to the verified PM2.5 signature: pollutant 88101, Micrograms/cubic meter (LC), FEM method 636. Compute a median per site/hour and then a median across available sites for that hour. This gives sites equal weight if monitor counts differ. Site counts and flagged-observation counts are retained; six sites are not hard-coded. Incompatible pollutant/unit/method combinations require a revised contract.

Weather and Air are left-joined on `pickup_hour_utc` only for trips whose pickup borough belongs to NYC's five boroughs. Missing hours, unknown locations and out-of-city pickups retain null environmental values. The result preserves Taxi fields and adds prefixed environmental fields, scope and match flags. Integration verifies both row count and unique record count before writing by `pickup_date`, then checks the written row count.

Verified data and testing

Full preparation produced 9,554,576 accepted Taxi rows and 202 rejected rows; Weather accepted 8,784, Air accepted 51,885 and Zones accepted 265. Air excluded 8,087,666 national rows. The integrated output contains 9,554,576 unique trips. Among 9,517,007 NYC-scope trips, weather and Air hour match rates are 100%; overall coverage is about 99.61%. Measurement completeness is reported separately, including missing snow and gust measurements.

Small Spark/Delta regression suites cover classification, malformed times, duplicate preference, special locations, cross-midnight/DST conversion, missing observations, incompatible Air signatures, join row preservation and Delta readback. Full-data verification complements these fixtures; it does not replace them.

Performance and limitations

Task 6 measures raw-to-Delta ingestion, current data bytes/file counts, and three required aggregations: trips per pickup borough, average duration per local date and average fare_amount per pickup borough. A one-week query additionally examines pruning. Layout order is reversed across ingestion rounds; queries are warmed, alternated, repeated and checked for equal results. See `docs/benchmark_report.md` for measurements and limitations, and `docs/architecture.md` for the architecture diagram.

NYC-wide environmental summaries are regional context, not observations at each street. Weather location/timezone remain assumptions; high matching rates cannot validate them. Same-hour aggregates may include observations after pickup and need a leakage review before prediction work. One local dataset scale cannot establish cluster or 20-times-volume scalability. Later weeks should separately evaluate file compaction, date layouts, caching, AQE and incremental processing.